MATH256
Linear Algebra HW#5

1. (3.1.6) Compute the determinants in this question using a cofactor expansion across the first row .

$$\begin{vmatrix} 5 & -2 & 4 \\ 0 & 3 & -5 \\ 2 & -4 & 7 \end{vmatrix}$$

2. (3.1.10) Compute the determinants by cofactor expansions . At each step , choose a row or column that involves the least amount of computation.

$$\begin{vmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -6 & -7 & 5 \\ 5 & 0 & 4 & 4 \end{vmatrix}$$

In questions 3.1.22 and 3.1.24 , explore the effect of an elemantery row operation on the determinant of a matrix. In each case , state the row operation and describe how it affects the determinant.

3. (3.1.22)

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}, \begin{bmatrix} a+kc & b+kd \\ c & d \end{bmatrix}$$

4. (3.1.24)

$$\begin{bmatrix} a & b & c \\ 3 & 2 & 2 \\ 6 & 5 & 6 \end{bmatrix}, \begin{bmatrix} 3 & 2 & 2 \\ a & b & c \\ 6 & 5 & 6 \end{bmatrix}$$

5. (3.2.8) Find the determinants by row reduction to echalon form

$$\begin{bmatrix} 1 & 3 & 3 & -4 \\ 0 & 1 & 2 & -5 \\ 2 & 5 & 4 & -3 \\ -3 & -7 & -5 & 2 \end{bmatrix}$$

6. (3.2.14)Combine the methods of row reduction and cofactor expansion to compute the determinants

$$\begin{bmatrix} -3 & -2 & 1 & -4 \\ 1 & 3 & 0 & -3 \\ -3 & 4 & -2 & 8 \\ 3 & -4 & 0 & 4 \end{bmatrix}$$

7. (3.3.6) Use the Cramer's rule to compute the solutions of the system

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 4 \\ -x_1 + 2x_3 &= 2 \\ 3x_1 + x_2 + 3x_3 &= -2 \end{aligned}$$